

In-Depth Review — Chapter 2 (Dataset)

Thesis: Binary classification of Raman spectra of biological tissue **File reviewed:**

Chapters/Ch_2_dataset.tex **Date:** 23 June 2026

Severity tiers used throughout: **[MUST FIX]**, **[SHOULD FIX]**, **[POLISH]**.

1. Overall assessment

The chapter has the right skeleton (origin/structure → labels → spectra → problem definition), and the reasoning in the *Binary classification* section is genuinely good — the trade-off analysis between the two strategies is the strongest part. But the writing quality is **well below Chapter 1**. Chapter 1 reads as finished, careful academic prose; Chapter 2 reads as working notes: telegraphic sentences, inconsistent tense, several broken LaTeX fragments, two literal placeholders (a blank (-----) and the PoliMi logo standing in for a figure), and a number of factual / figure-caption inconsistencies.

The single biggest content gap is that **this is a dataset chapter for an ML thesis that barely describes data preprocessing** — for Raman spectroscopy that is a central methodological topic, and right now only "we cropped the silent region" is covered.

2. Critical issues — MUST FIX

- Blank placeholder in the text (line 96).** "originally acquired in the range (-----)" — the acquisition range is missing entirely. This is the one number a reader needs to interpret every spectrum figure. Fill it (the full measured Raman-shift range in cm^{-1}).
- "14 classes" but only 13 are shown.** Line 76 says "divided in 14 classes"; `label_distributions.png` lists exactly 13 (Tumor-2, Tumor stroma-6, Stroma w/ inflammation-4, Connective-5, Tumor+necrosis-20, Unlabeled-(-1), Artifact-15, Epithelium-19, Skeletal muscle-3, Necrosis-0, Glandular-8, Blood vessel-10, Lymphatic-23). Either a 14th class has zero counts (then say so) or the number is wrong.
- Patient vs sample contradiction.** Line 11 says 25 samples "each one from a different patient" ($\Rightarrow$ 25 patients), but the `Sample_ID` format `1234_567` is explicitly "first 4 digits = patient, last 3 = sample from that patient" (line 24) — which only makes sense if a patient can contribute several samples. State clearly: how many patients, how many samples, and whether any patient appears more than once. This matters later for train/test splitting (leakage).
- The organ / cancer type is never stated.** The whole thesis is about classifying tumour tissue and you never say *what tissue* (breast? colon? skin? head-and-neck?). This is essential context and a reviewer will ask immediately.

5. **Figure caption says "discarded classes" but necrosis is not discarded.**

`fig:spectra_comparison_discarded` (caption, line 114) compares tumor against Necrosis(0), Artifact(15), Unlabeled(-1). Only artifact and transparent are actually removed; necrosis is kept (it appears in the distribution and is discussed as a possible member of the tumor macro-class, line 150). Calling all three "discarded classes" is inaccurate and confusing, especially since the *same* figure is reused in two sections for two different arguments (line 79 vs line 109).

6. **The figure does not strongly support the claim built on it.** Lines 79 and 109 say the discarded/necrosis spectra "deviate from all tissue classes" / "noticeably differed from all others." In `spectra_comparison_discarded.png`, all four curves track each other closely (same fingerprint peaks, same dominant CH peak $\approx$ band 420); the deviations are subtle, not "noticeable." Either soften the wording, point to the *specific* regions where they differ, or add a quantitative measure (e.g. spectral distance / correlation to the tumor mean).

7. **Broken LaTeX, line 154:** `\textit{connective tissue}(5))` — stray/extra parenthesis and missing space; the whole `i.e. ...` list is malformed. Also `(>1000000 spectra)` should be `(>\$1\,000\,000 spectra)` (math `>`, thin-space grouping).

8. **Subfloat caption typos (lines 169, 173):** `(1st approach.` and `(2nd approach.` — missing closing `)`. And line 176 `choiche` → `choice`.

3. Section-by-section walkthrough

Chapter intro (line 4)

- **[SHOULD FIX]** "This chapter explains how the dataset is structured." is too thin and undersells the chapter (it also covers labels, spectra, and the binary problem definition). Expand into a 2–3 sentence preview of all four sections, mirroring the closing-paragraph style of Chapter 1.

§2.1 Origin and Structure

- **[POLISH]** Line 13: don't start a sentence with a numeral — "One to five areas, called *maps*..."
- **[SHOULD FIX]** Line 14: "120x120" → `120{\times}120` (and everywhere "x" means dimensions). Same for pixel ranges.
- **[SHOULD FIX]** Lines 15 / 34–35: state the **spectral resolution** (cm^{-1} per band) and the actual wavenumber axis, not just "483 bands." Right now the reader cannot map a "band index" in any figure to a cm^{-1} value. Connects to issue #1.
- **[POLISH]** Line 16 / line 58: "Each spectrum (i.e. pixel)" vs "each pixel (spectrum)" — fine, but use `i.e.\` (escaped space) to avoid LaTeX inter-sentence spacing.
- **[POLISH]** Line 44: good arithmetic ($78 \times 120^2 = 1,123,200$ ✓). Consider also giving the **cleaned** count (after removing artifact + transparent) and the **working** count after subsampling.
- **[MUST FIX]** Lines 48–51: the figure is the PoliMi logo as a placeholder (you flagged this). The intended sample→map→spectrum schematic would genuinely help. Caption "Relation between Sample, maps and spectra" also has inconsistent capitalization → "Relationship between samples, maps and spectra." Rename `fig:placeholder` to something meaningful.

- **[POLISH]** A **table mapping label integer** → **tissue name** → **count** → % would be more useful (and citable) than reading numbers off the bar chart. You already have all the data.

§2.2 Labels

- The annotation pipeline (12 μm slices, Raman on one slice, H&E on the adjacent slice, pathologist annotation transferred back) is well explained, and the three "critical points" are a nice touch.
- **[SHOULD FIX]** Line 69: "Human classification is not pixel-wise, as is artificial classification" is hard to parse. Rephrase: "Unlike automated, pixel-wise classification, human annotation delineates regions coarsely; errors are therefore concentrated at tissue boundaries."
- **[POLISH]** Lines 67–71: inconsistent end punctuation across the three bullets. Make uniform.
- **[SHOULD FIX]** Line 73: "as no other mean of comparison are available" → "as no other means of comparison **is** available." Consider softening: the labels are treated as ground truth *despite* known limitations — say it that way.
- **[SHOULD FIX]** The term "critical points" (here and line 148) reads as a calque from Italian. "Potential sources of error" / "limitations" is clearer.
- **[MUST/SHOULD FIX]** Line 76: "47,92%" — comma decimal separator (Italian). Use "47.92%". Also reconcile with the "47%" used at line 154.
- **[POLISH]** Line 76: missing period after `\Cref{fig:label_distribution}`.
- **[POLISH]** Lines 81–92: the two subfigures sit at very different scales/aspect ratios; the binary one is mostly whitespace. Consider trimming/standardizing.

§2.3 Spectra

- **[MUST FIX]** Line 96: the blank range (#1) and "reduce the training data weight by 30%". "Data weight" is informal → "reduce the data volume by ~30%." Also make explicit whether the **483 bands already exclude** the silent region (if 30% was removed, the original must have been ≈ 690 bands — but line 15 calls 483 the spectrum length; currently ambiguous).
- **[SHOULD FIX]** Cross-chapter inconsistency: Chapter 1 (line 120) puts the C–H/high-wavenumber region at **2800–3000 cm^{-1}** ; here (line 109) you write the "CH region (2800–3500)". Pick one and make Ch.1 and Ch.2 agree.
- **[MUST FIX]** **Normalization is undefined.** The comparison captions (line 121) say "normalized" and those plots are clearly standardized (mean ≈ 0 , range $-2\dots 4$), while `spectrum_avg_tumor.png` is in raw intensity (0–0.2). Different normalizations appear across figures with no explanation of which method (vector norm? SNV? area? z-score?) was used where or why. For a Raman ML thesis this must be specified.
- **[MUST FIX]** **Preprocessing pipeline is essentially absent.** No mention of cosmic-ray removal, baseline/fluorescence-background subtraction, smoothing/denoising, or outlier rejection — all standard and all relevant to "the model may learn noise" (Ch.1 line 155 explicitly warns about this). At minimum, state what was and wasn't done.
- **[SHOULD FIX]** Lines 99 & 109: figures use "**Band Index**" on the x-axis, but the text discusses "bands 30–200", "500–1800 cm^{-1} ", "2800–3500". The reader can't reconcile band index with cm^{-1} . Relabel the axes in cm^{-1} (or provide the conversion).
- **[POLISH]** Line 104 caption: "In light blue the standard deviation" is a fragment → "The shaded band shows ± 1 standard deviation across spectra."

- **[POLISH]** Line 99: "standard deviation ... corresponds to 12–20% of the peak height" — good, quantitative. Tie the "bands 30–200" claim to a cm^{-1} range.
- **[POLISH]** Tense drift: "All classes **appear** ... The only exceptions are ... which noticeably **differed**" (line 109). Pick one tense.
- **[POLISH]** Line 109: missing period after the `\Cref` .

§2.4 Binary classification

- **[SHOULD FIX] Title is misleading.** No classification happens here — this section *defines the binary problem*. Rename to "Definition of the binary classification problem" or "Binary labelling strategy."
 - The two-strategy analysis and the three criteria (relevance / numerosity / discernability) are the best-written content in the chapter. Keep it; just tighten.
 - **[SHOULD FIX]** Line 150: "necrotic tissue ... show high variance" → "shows." Naming: text says "tumor+necrosis (20)" but the figure says "Tumor with necrosis – 20." Standardize one spelling everywhere.
 - **[MUST FIX]** Line 154: malformed list (#7) and the unjustified quantitative claim "subsampling ... without major shortcomings down to 5–10%." That robustness claim (and the "adjacent pixels are highly similar / spatial autocorrelation" rationale) should be supported — a citation, or your own correlation evidence.
 - **[SHOULD FIX]** Line 161: "classified based **of** the histopathologist's judgement" → "based **on**."
 - **[SHOULD FIX]** Line 165 is the **most important sentence in the chapter** (the actual decision) and it's grammatically broken: "...while the Tumor vs. Tumor stroma as a surgically oriented problem... expecting for it better results overall." Rewrite as a clean statement, ideally a short two-item summary:
 - *Problem A — Tumor vs. Healthy*: all valid non-tumor classes pooled; tests the global, clinically versatile question; $\approx 50/50$ balance.
 - *Problem B — Tumor vs. Tumor stroma*: spectrally homogeneous, surgically motivated (margin assessment); requires subsampling tumor for balance.
 - **[SHOULD FIX]** Buried-but-important point (line 176): "No map in the dataset contained both tumor stroma and other labels." That means tumor-stroma maps are *pure*, which is why you had to illustrate with stroma-with-inflammation instead. This has real consequences for train/test independence and deserves a sentence in the body, not just a caption aside.
 - **[POLISH]** Line 176 caption is a run-on with a dangling clause and "choiche" typo; rewrite: "*Stroma with inflammation* is shown instead of *tumor stroma* (the actual training target) for illustration, because no map contained *tumor stroma* together with other labels."
 - **[POLISH]** Heatmap figures: titles include the literal sample ID "3260_005"; say in the caption that it is a single representative sample.
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4. Typos, grammar & LaTeX nits (consolidated)

Line	Issue	Fix
4	Thin intro	Expand to preview all sections
13	Sentence starts with numeral	"One to five areas..."
14	120x120	120{\times}120
49	"Sample, maps and spectra" caps	"samples, maps and spectra"
73	"no other mean ... are"	"no other means ... is"
76	"47,92%"	"47.92%"
76, 109	Missing terminal periods after \Cref	add
96	(-----) ; "training data weight"	fill range; "data volume"
96, 109	straight quotes 'silent region'	LaTeX quotes "silent region" ``
104	caption fragment	full sentence
150	"show"	"shows"
154	connective tissue}(5)), >1000000	fix parens; \$>\$1\,000\,000
156	\cref lowercase	\Cref (consistency)
161	"based of"	"based on"
165	broken sentence	rewrite (see above)
169, 173	(1st approach. / (2nd approach.	add)
170, 174	Images//Images_ch_2 double slash	single
176	"choiche", dangling clause	"choice"; rewrite

Cross-cutting consistency to enforce throughout:

- **Tense:** choose past (data/methods) consistently; you currently mix past/present.
- **Class-name italics:** sometimes `\textit{tumor}`, sometimes plain — make uniform.
- **Class naming:** "tumor+necrosis" vs "Tumor with necrosis"; "tumor stroma" vs "Tumor stroma" capitalization.
- **\cref vs \Cref** : standardize.
- **Numbers:** period decimals; \, thousands grouping everywhere (1\,000\,000).
- **No citations at all** in the chapter — at least the H&E annotation-transfer method, the silent-region rationale, and the subsampling-robustness claim should be referenced.

5. Missing content / suggested additions

1. **Preprocessing section** (cosmic-ray removal, baseline correction, normalization method + rationale, smoothing/denoising, outlier handling). Currently the most serious gap.

2. **Acquisition range + spectral resolution in cm^{-1}** , and relabel figure x-axes in cm^{-1} .
 3. **Class table**: integer label $\rightarrow$ tissue name $\rightarrow$ description $\rightarrow$ count $\rightarrow$ %.
 4. **Organ/cancer type** and basic provenance.
 5. **Ethics / data-provenance sentence** (human tissue: approval, anonymization, source institution) — standard and usually expected.
 6. **Patients-vs-samples clarification** + a short note (or histogram) of **maps per sample**.
 7. **Explicit data-independence statement**: splits will be done at patient/sample level to prevent leakage from spatially adjacent, near-identical pixels. You hint at it ("keeping spectra from each sample") but never state the principle.
 8. **Working-set sizes**: spectra remaining after cleaning, and after subsampling for each binary problem.
 9. Replace the **logo placeholder** with the sample $\rightarrow$ map $\rightarrow$ spectrum schematic (and optionally a small diagram of the H&E adjacent-slice annotation transfer).
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6. Structural / "thesis quality" notes

- **Bring the prose up to Chapter 1's register.** There is a visible quality cliff. Many sentences are note-like; add connective tissue between sections (each section currently starts cold).
 - **Consider renaming the chapter** "Dataset and Problem Definition" (it does more than describe the dataset) and renaming §2.4.
 - **Figure reuse:** `spectra_comparison_discarded` doing double duty for two distinct arguments is confusing; either split into two purpose-built figures or make one caption cover both uses explicitly.
 - **The decision logic is sound but the payoff is hidden.** The chapter's whole point — *which* binary problems you'll attack and *why* — lands in one broken sentence (165). Give it a clear, prominent closing paragraph; it is the bridge to the ML chapters.
 - **Forward references:** since PCA/ANN chapters are coming, add one-line "as discussed in Chapter X" pointers where you defer (subsampling, splitting, normalization) so the reader knows it is intentional, not omitted.
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7. Highest-value quick wins (do these first)

1. Fill the blank acquisition range (line 96) and add a cm^{-1} axis.
2. Add a preprocessing/normalization paragraph.
3. Fix the 14-vs-13 class count and the patient/sample contradiction; name the organ.
4. Repair the broken LaTeX / captions (lines 154, 165, 169, 173, 176).
5. Rewrite the decision sentence (165) into a clear two-problem summary.
6. Fix the "discarded classes" caption / over-strong "deviate from all classes" claim.